SpaceX Falcon 9 First Stage Landing Prediction

Data Science Capstone Project Report

IBM Applied Data Science Capstone

Magaly del Carmen Fonseca Medrano | November 2026

Executive Summary

Methods used

- Data collection: SpaceX API and web scraping
- Data wrangling and EDA with visuals and SQL
- Interactive analytics: Folium map and Plotly Dash
- Predictive models for landing success

Key results

- Launch site with the most successes: KSC LC-39A (about 42% of all successful launches)
- Success rate trend over the years: rising, from no successful landings before 2014 to about 85% to 90% in recent years
- Best model: Decision Tree with 83.33% accuracy
- Main drivers of success: launch site, payload mass (about 2,000 to 5,500 kg) and orbit

Introduction

Background

- Falcon 9 launch costs about $62 million
- Other providers charge $165 million or more
- Savings come from reusing the first stage

The problem

- Predict if the first stage will land
- Landing outcome sets the launch cost
- Goal: help a competitor bid against SpaceX

Questions

- Which factors drive landing success?
- How do site, payload and orbit matter?
- Which model predicts best?

Data Collection: SpaceX API



requests.get()
api.spacexdata.com/v4

json_normalize()

rocket, launchpad,
payload, cores

keep Falcon 9 only

fill missing payloads
export CSV

Output: one row per Falcon 9 launch with site, payload, orbit and landing result

GitHub URL: https://github.com/mfm2174/Coursera-Data-Science.git

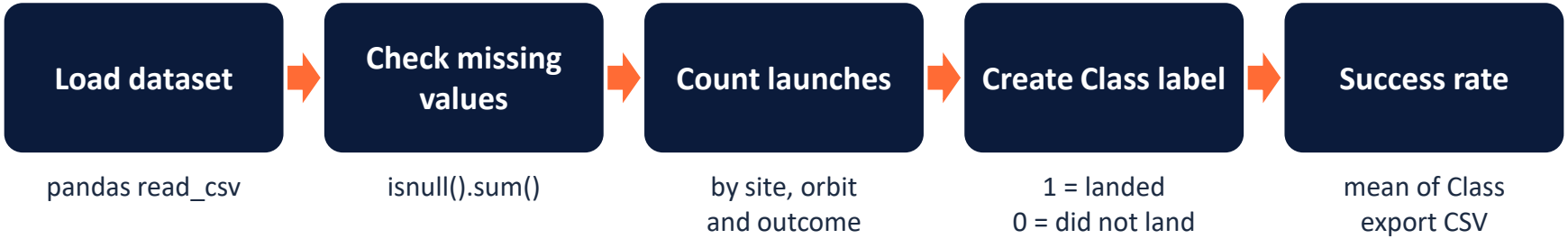
Data Collection: Web Scraping



Output: launch table with date, booster version, launch site and outcome

GitHub URL: https://github.com/mfm2174/Coursera-Data-Science.git

Data Wrangling Methodology



Output: clean dataset ready for EDA and machine learning

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EDA with Data Visualization

Scatter plots

- Flight number, payload, site, orbit vs outcome
- Purpose: spot relationships

Bar chart

- Success rate by orbit
- Purpose: compare categories

Line chart

- Yearly success trend
- Purpose: see change over time

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EDA with SQL

Launch sites and payloads

- Unique launch sites
- Sites starting with 'CCA'
- Total payload for NASA boosters
- Average payload for F9 v1.1

Outcomes, rankings and time

- First successful ground pad landing
- Successful drone ship, payload 4000 to 6000
- Total successful vs failed missions
- Landing outcome ranking by date

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Interactive Visual Analytics

Folium map

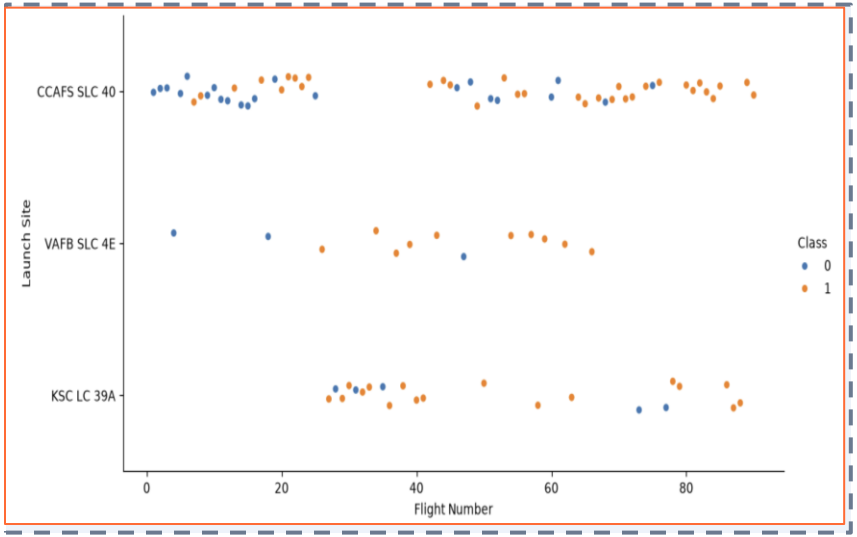
- Markers for every launch site
- Green and red markers for success and failure
- Distance lines to coast, railway, highway and city

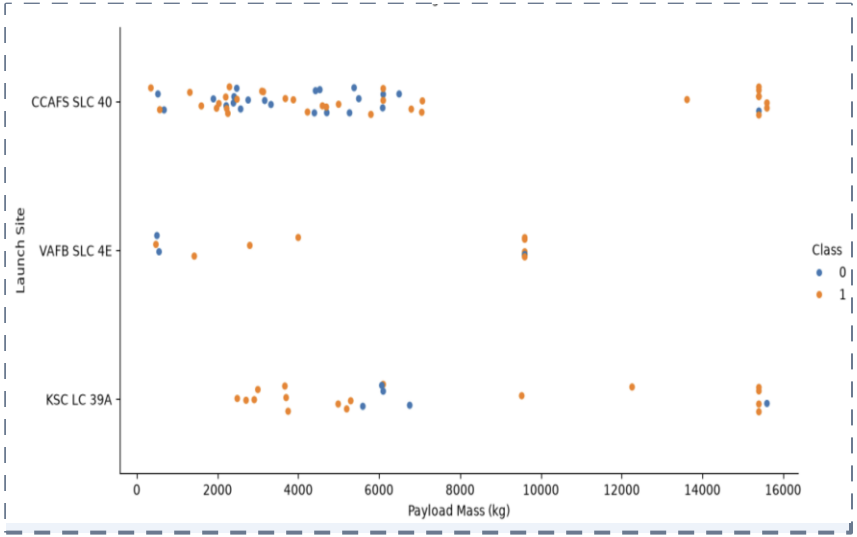
Plotly Dash app

- Launch site dropdown
- Success pie chart
- Payload slider and payload vs outcome scatter

GitHub URLs:
Folium: https://github.com/mfm2174/Coursera-Data-Science.git| Dash: https://github.com/mfm2174/Coursera-Data-Science.git

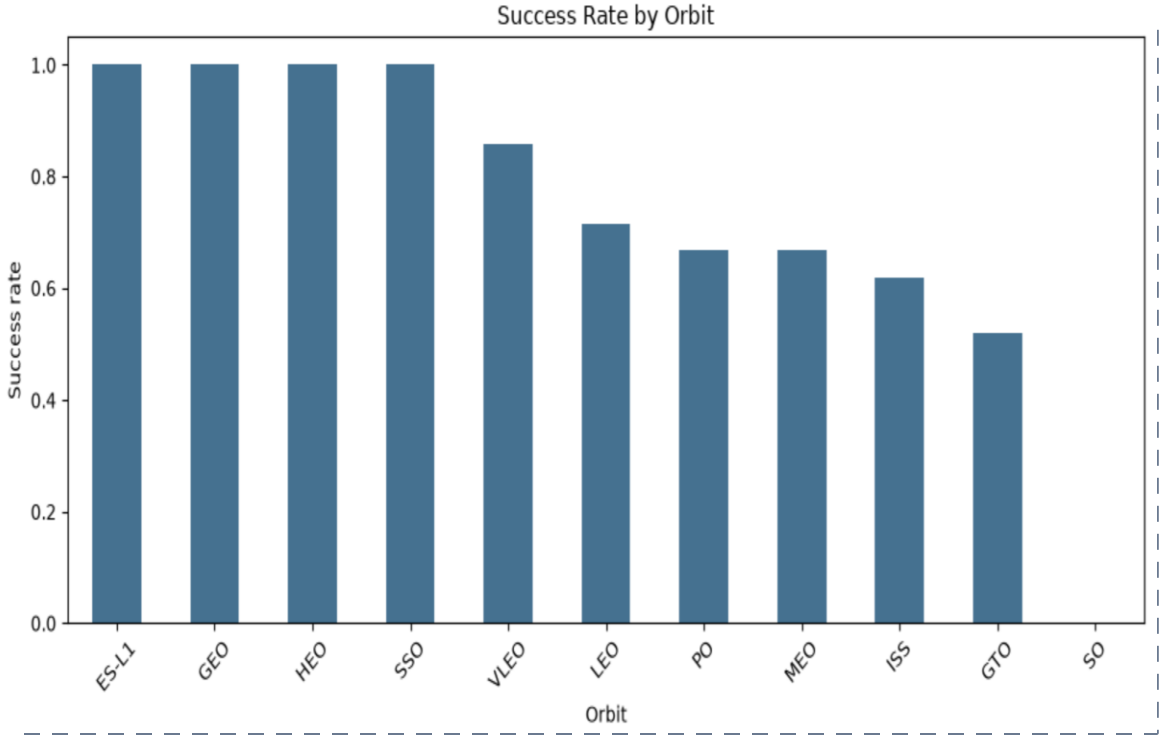
EDA Results: Scatter Plots





Insight: Success improves over time and with heavier payloads; KSC LC 39A and later CCAFS flights are almost all successful.

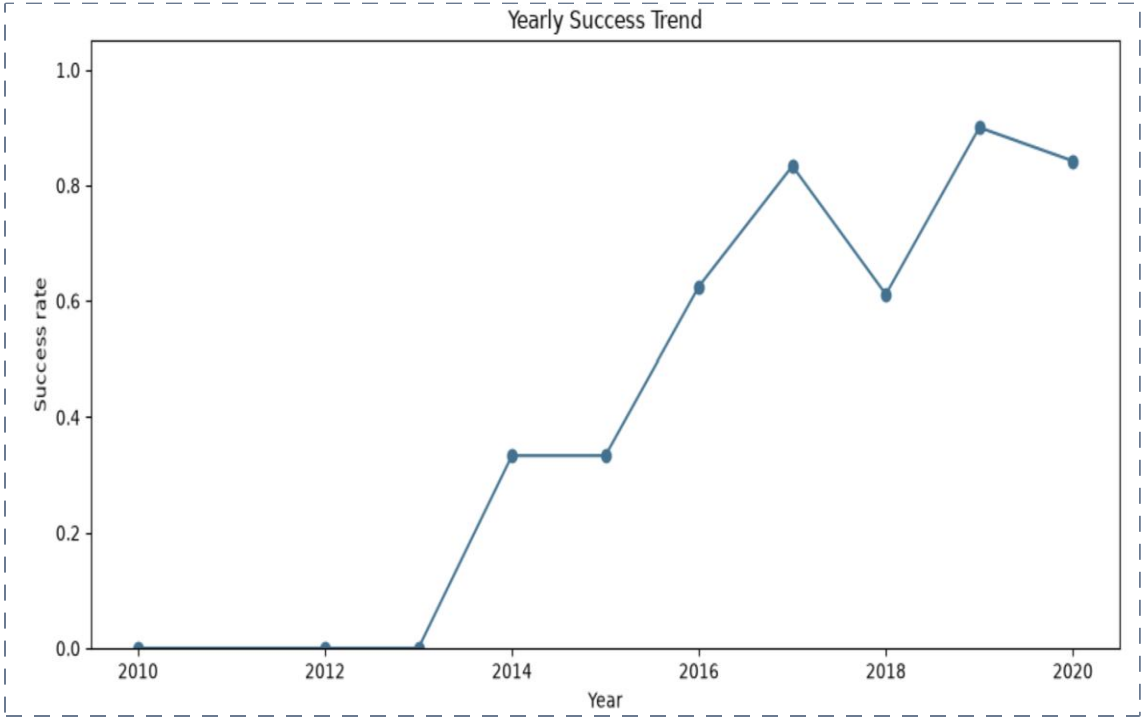
EDA Results: Success Rate by Orbit



What it shows

- Orbits with the highest rate: ES-L1, GEO, HEO and SSO
- Orbits with the lowest rate: SO
- Why it matters: Demanding orbits like GTO have the lowest success among common orbits, so orbit is a key input for the prediction model.

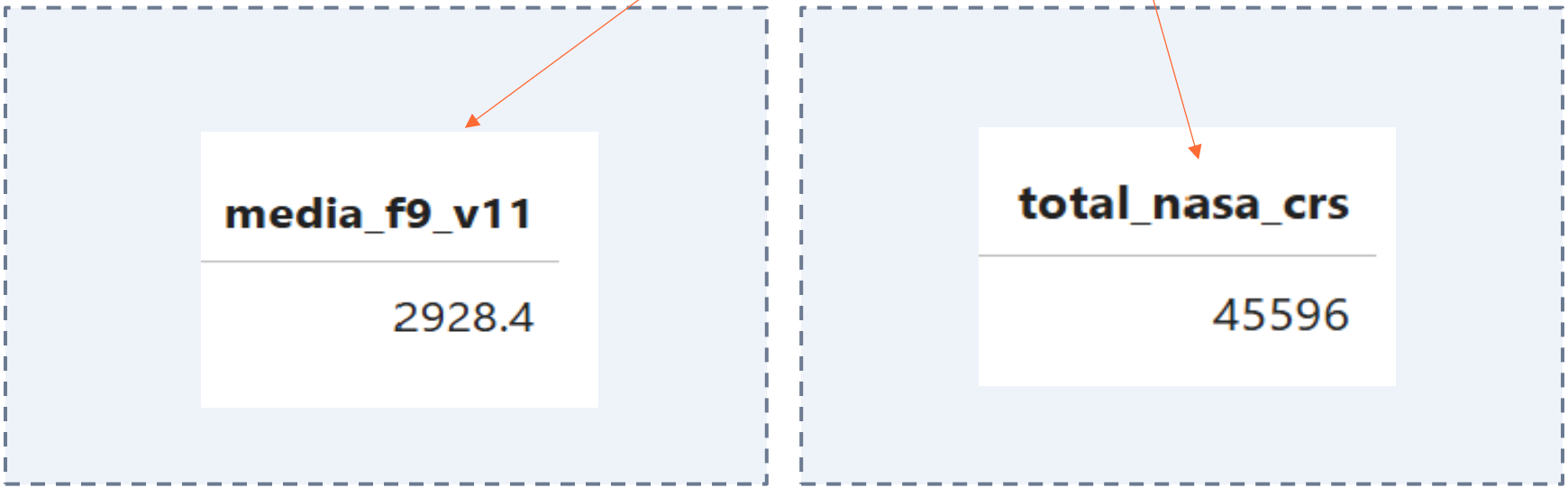
EDA Results: Yearly Success Trend



What it shows

- Trend: Success rate rises over the years, with a dip in 2018
- Turning point year: 2014
- Why it matters: Each launch teaches SpaceX something, so recent launches are more reliable

SQL Results: Launch Sites and Payloads



Insight: Launches happen at 4 sites, NASA (CRS) missions carried 45,596 kg in total, and the F9 v1.1 averaged about 2,928 kg per launch.

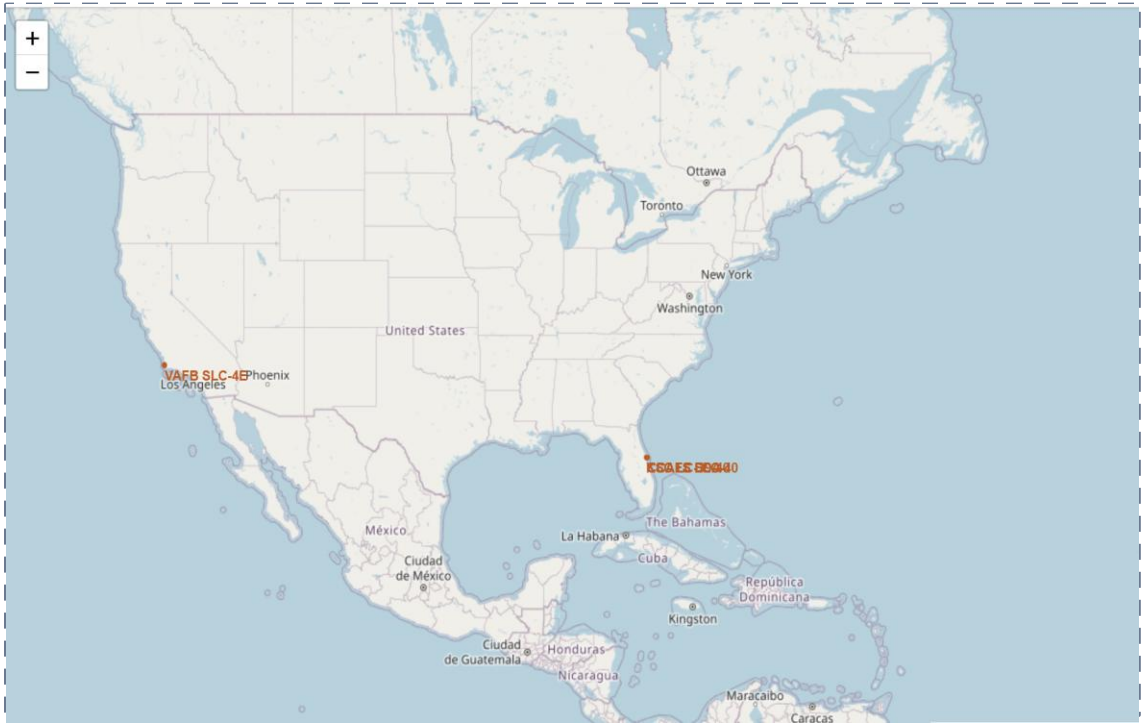
SQL Results: Success, Rankings and Time

	Mission_Outcome	total
0	Failure (in flight)	1
1	Success	99
2	Success (payload status unclear)	1

	Landing_Outcome	total
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

Insight: Almost all missions succeed (100 of 101), and in the early years most launches made no landing attempt, with drone ship landings split evenly between success and failure.

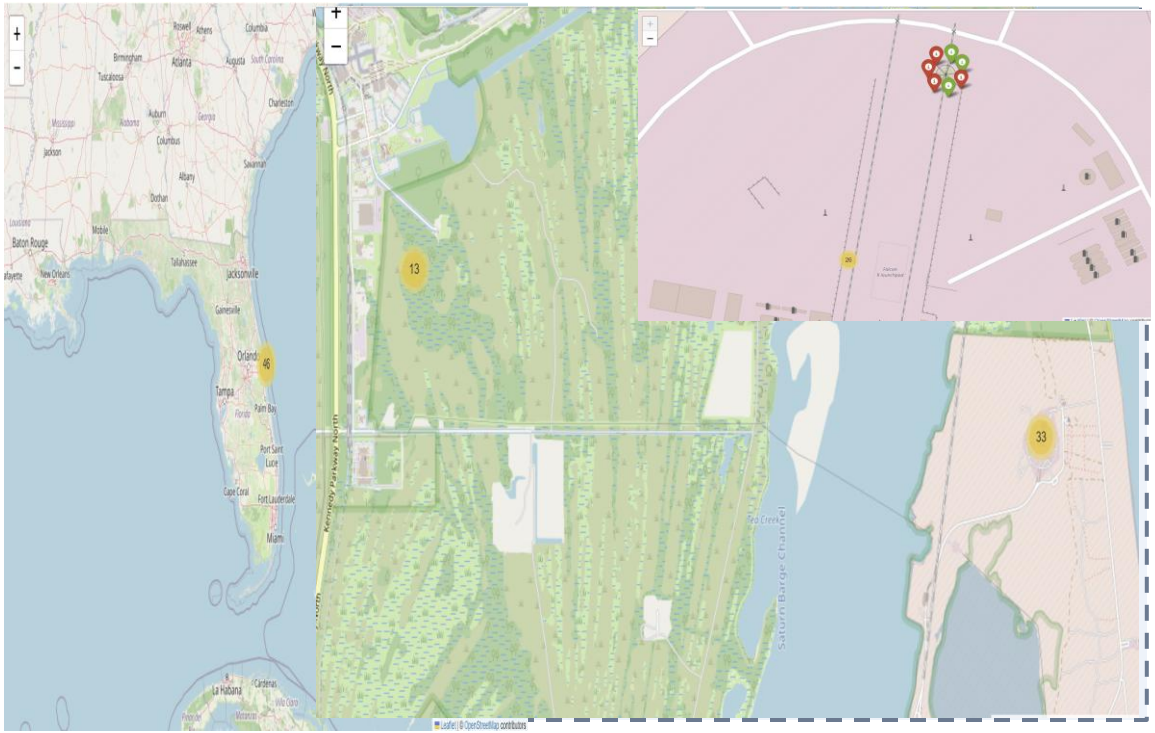
Folium Map: Launch Site Markers



What it shows

- Each site has a circle and label
- Location pattern: All sites are close to the coast: three in Florida and one in California
- Why it matters: Launching near the coast keeps rockets over the ocean, away from populated areas

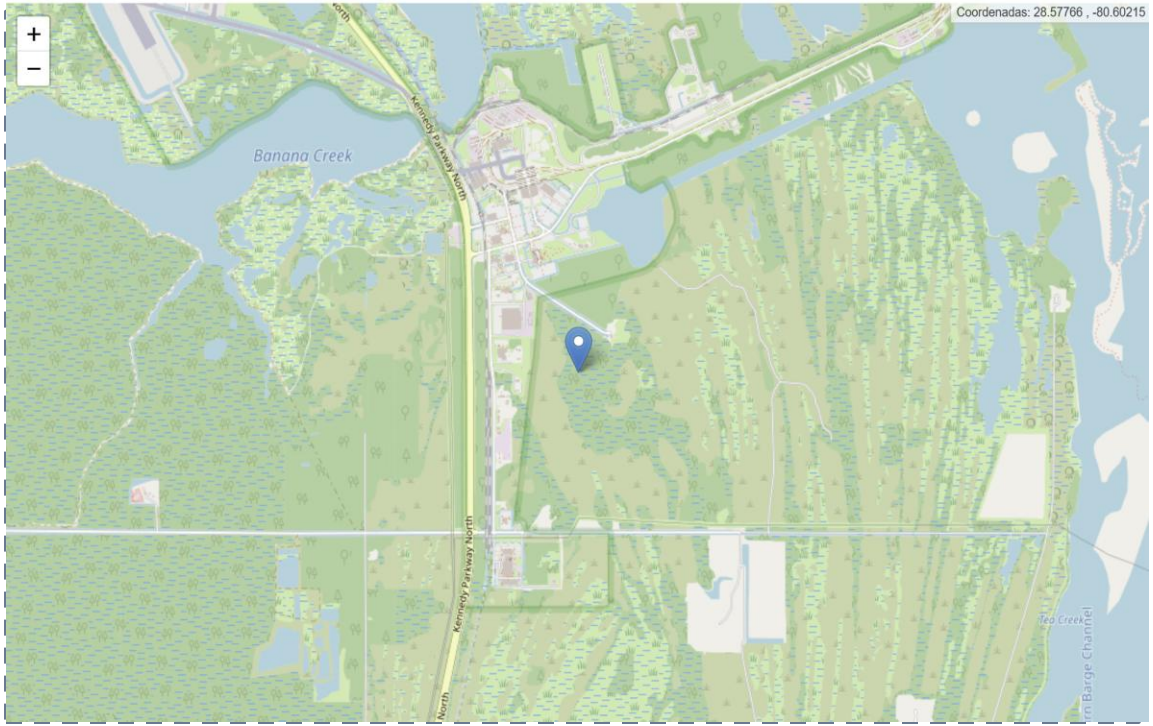
Folium Map: Launch Records



What it shows

- Green = successful landing
- Red = failed landing
- Best site: KSC LC-39A

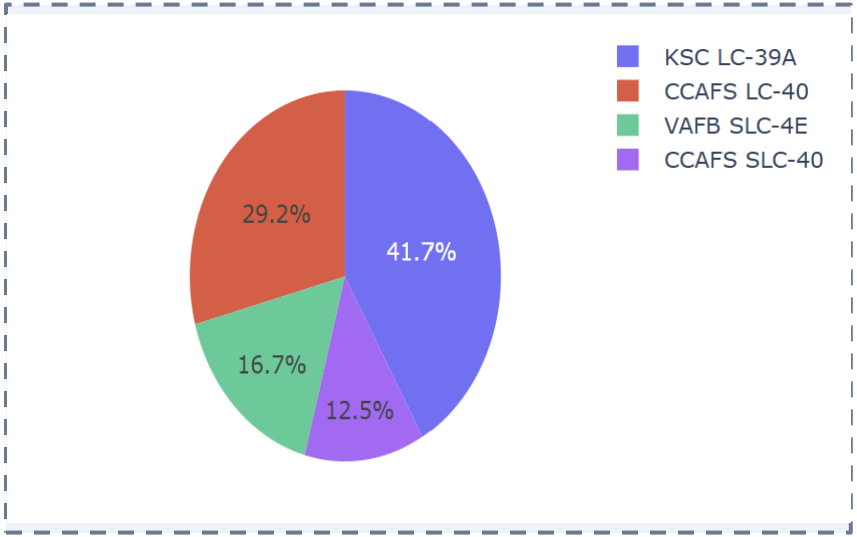
Folium Map: Proximity Analysis

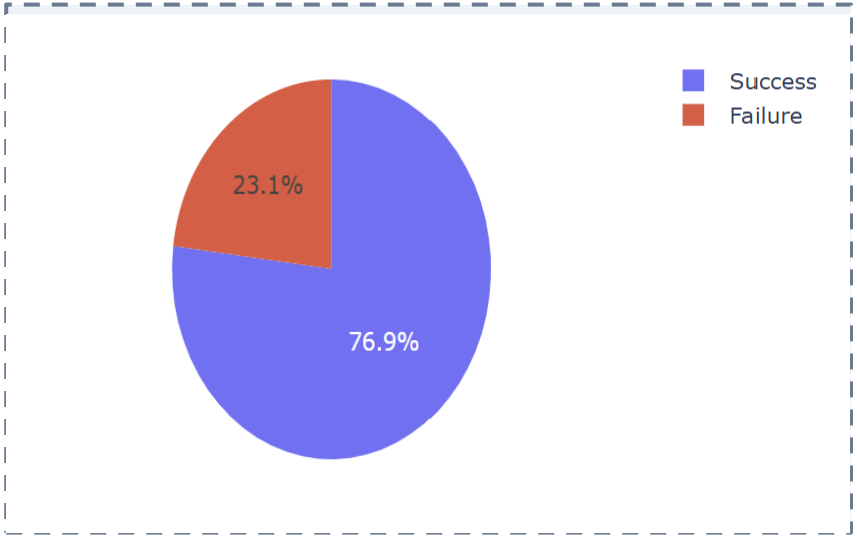


What it shows

- Distance to coast: 4.81 km
- Distance to railway and highway: 9097.98 km
- Distance to city: 9097.98 km

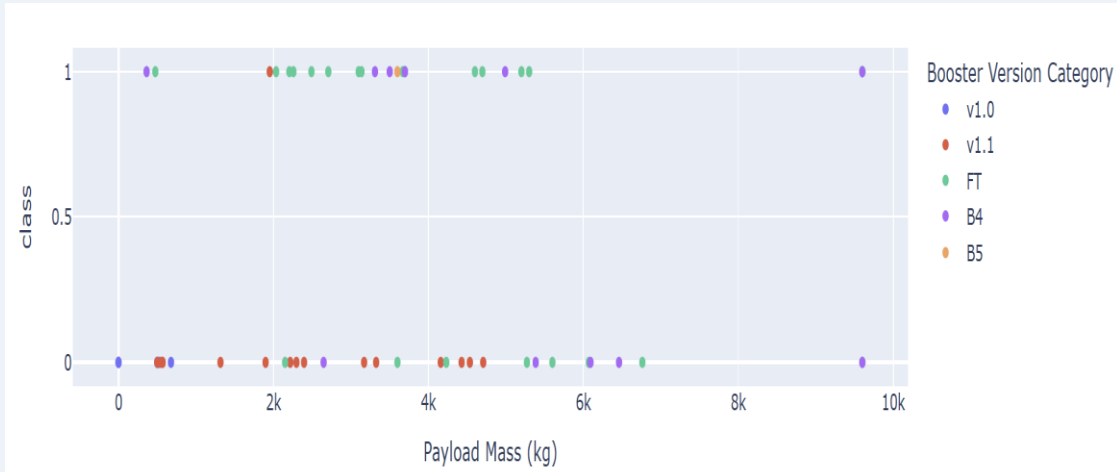
Plotly Dash: Success Pie Charts





Insight: KSC LC-39A leads with about 42% of all successful launches, about 12 points ahead of the second site, and about 77% of its own launches succeed.

Plotly Dash: Payload vs Launch Outcome



What it shows

- Payload range slider
- Color = booster version
- Best payload range: about 2,000 to 5,500 kg, mostly with the FT booster

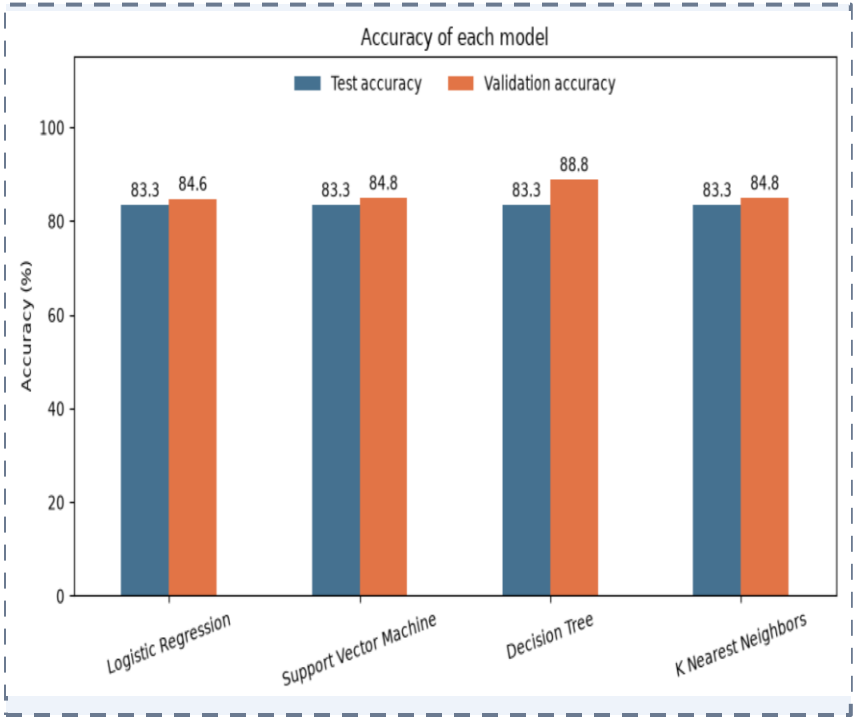
Predictive Analysis: Models



Target: Class = 1 if the first stage landed, 0 if it did not

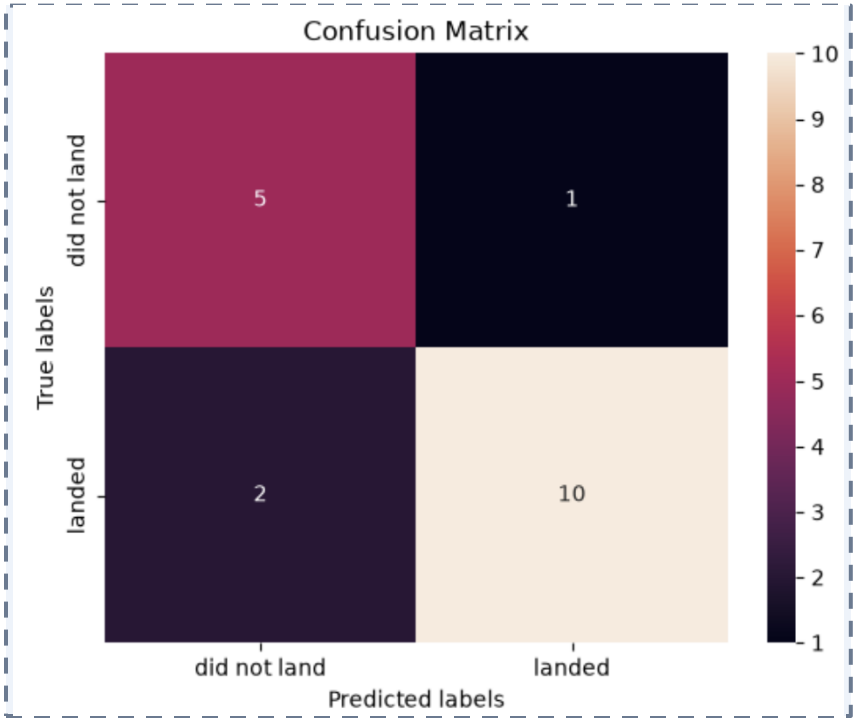
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Model Evaluation Results



Model	Test accuracy
Logistic Regression	84.6%
Support Vector Machine	84.8%
Decision Tree	88.8%
K Nearest Neighbors	84.8%

Confusion Matrix and Best Model



Best model: [add name]

- Accuracy: 83.33%
- Why it won: highest validation accuracy (88.75%), tied on the test set
- Reads well when it predicts landings
- Main mistake: it misses some landings that actually succeed (false negatives)

Conclusion and Insights

Key findings

- Launch site: KSC LC-39A has the best record, with about 77% successful landings
- Payload and orbit: landings succeed most with payloads of about 2,000 to 5,500 kg, and GTO is the hardest common orbit
- Success improves over time: from no successful landings before 2014 to about 85% to 90% in recent years
- Best model: Decision Tree, with 83.33% accuracy on the test set

Creative insights

- Use the model to predict a landing before launch, and turn it into a launch cost estimate
- Prefer KSC LC-39A and payloads of about 2,000 to 5,500 kg, where landings succeed most
- Add weather and booster reuse data, and more launches, to make the model more reliable
- Share the Dash app as a tool to compare launch scenarios and support bidding against SpaceX